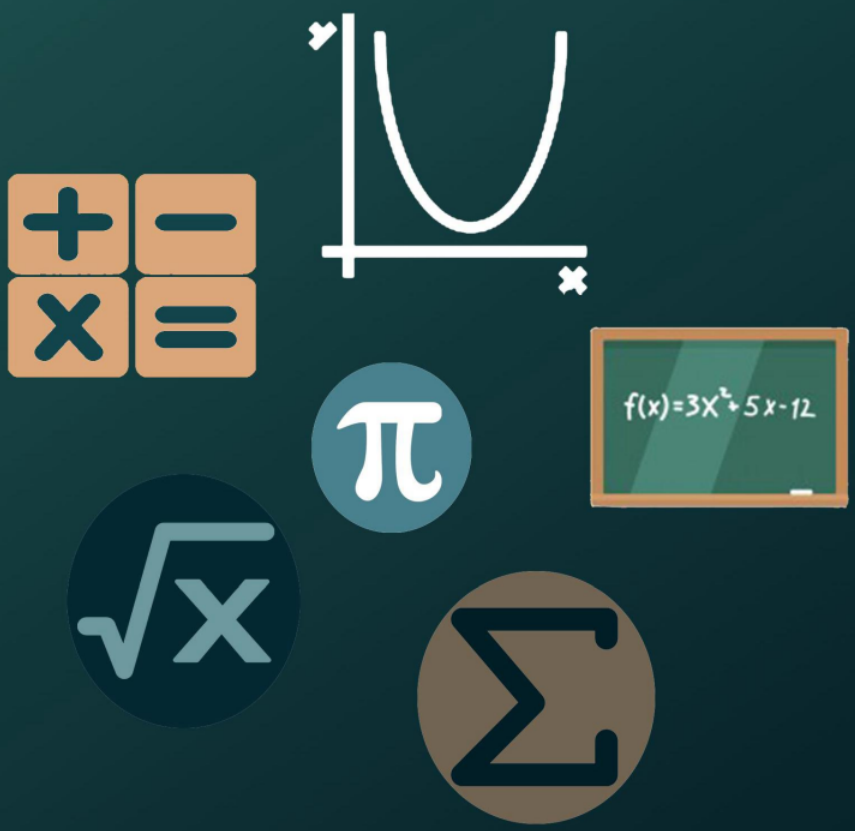
MATHEMATICS

Ellipse



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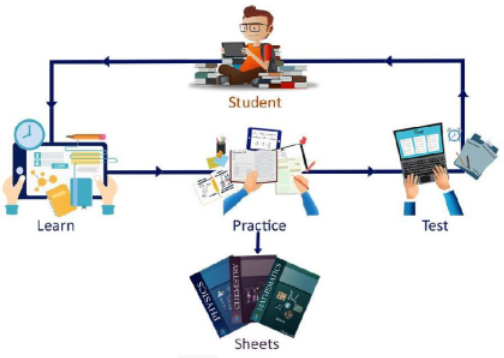
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Standard Ellipse				
2	Ellipse at a Glance				
3	Problems on Standard Ellipse	1 - 7	1	1-2	1
4	Other Forms of Ellipse	8 - 9	7-8	3-5	2, 7-9
5	Auxilliary Circle and Eccentricity of Ellipse	10-11		6-10	20
6	Point and Ellipse	12-13	9	11	
7	Tangent to Ellipse - Slope Form	14	10-12	12-13	21
8	Tangent to Ellipse - Cartesian Form	15	2-5, 13		10-13
9	Tangent to Ellipse - Parametric Form			14	22-25
10	Normals to Ellipse	16-18	14	15-16	26
11	Pair of Tangents to Ellipse	19		17	17
12	Chord with a given Mid-Point	20		18	18
13	After Complete Chapter		6	19	3-6, 14-16, 19, 27-29

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Problems on Standard Ellipse

Q1. In an ellipse, the distance between its foci is 6 and minor axis is 8. Then its eccentricity is-

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$

(C) $\frac{1}{\sqrt{5}}$ (D) $\frac{3}{5}$

Q2. The equation of an ellipse whose focus is (–1, 1) whose directrix is $x - y + 3 = 0$ and whose eccentricity is $\frac{1}{2}$, is given by :

(A) $7x^2 + 2xy + 7y^2 + 10x - 10y + 7 = 0$

(B) $7x^2 - 2xy + 7y^2 - 10x + 10y + 7 = 0$

(C) $7x^2 - 2xy + 7y^2 - 10x - 10y - 7 = 0$

(D) $7x^2 - 2xy + 7y^2 + 10x + 10y - 7 = 0$

Q3. The equation of the ellipse whose latus rectum is 8 and whose eccentricity is $\frac{1}{\sqrt{2}}$, referred to the principal axes of co-ordinates, is :

(A) $\frac{x^2}{8} + \frac{y^2}{9} = 1$ (B) $\frac{x^2}{64} + \frac{y^2}{32} = 1$

(C) $\frac{x^2}{16} + \frac{y^2}{24} = 1$ (D) $\frac{x^2}{18} + \frac{y^2}{32} = 1$

Q4. The length of the latus rectum of the ellipse $9x^2 + 4y^2 = 1$, is :

(A) $\frac{8}{9}$ (B) $\frac{8}{3}$ (C) $\frac{4}{9}$ (D) $\frac{3}{2}$

Q5. The eccentricity of the ellipse $4x^2 + 9y^2 = 36$, is :

(A) $\frac{\sqrt{5}}{3}$ (B) $\frac{1}{2\sqrt{3}}$

(C) $\frac{\sqrt{5}}{6}$ (D) $\frac{1}{\sqrt{3}}$

Q6. Equation of the ellipse with eccentricity $\frac{1}{2}$ and foci at $(\pm 1, 0)$ is :

(A) $\frac{x^2}{3} + \frac{y^2}{4} = 1$ (B) $\frac{x^2}{4} + \frac{y^2}{3} = 1$

(C) $\frac{x^2}{4} + \frac{y^2}{3} = \frac{4}{3}$ (D) None of these

Q7. The sum of focal distances of any point on the ellipse with major and minor axes as $2a$ and $2b$ respectivley is equal to :

(A) 2a (B) $\frac{2a}{b}$ (C) $\frac{2b}{a}$ (D) $\frac{b^2}{a}$

Other Forms of Ellipse

Q8. If the ellipse $\frac{(x-h)^2}{M} + \frac{(y-k)^2}{N} = 1$ has major axis on the line y = 2, minor axis on the line x = –1, major axis has length 10 and minor axis has length 4.

Then h + k + M + N is equal to -

(A) 10 (B) 15 (C) 27 (D) 30

Q9. The focii of the ellipse $25(x+1)^2+9(y+2)^2=225$ are :

(A) (1, –2) and (1, –6)

(B) (–1, 2) and (6, 1)

(C) (–1, –2) and (1, 6)

(D) (–1, 2) and (–1, –6)

Q.10 The equation of the circle passing through the points of intersection of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$ is -

(A) $x^2+y^2=a^2$

(B) $x^2+y^2=b^2$

(C) $x^2+y^2=\frac{a^2b^2}{a^2+b^2}$

(D) $x^2+y^2=\frac{2a^2b^2}{a^2+b^2}$

Q.11 Let S(5,12) and S'(–12,5) are the foci of an ellipse passing through the origin. The eccentricity of ellipse equals -

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{2}{3}$

Points and Ellipse

Q12. The number of real tangents that can be drawn to the ellipse $3x^2+5y^2=32$ or $25x^2+9y^2=450$ passing through (3, 5) is-

(A) 0 (B) 2

(C) 3 (D) 4

Q13. Let 'E' be the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ & 'C' be the circle $x^2+y^2=9$.

Let P & Q be the points (1 , 2) and (2, 1) respectively. Then :

(A) Q lies inside C but outside E

(B) Q lies outside both C & E

(C) P lies inside both C & E

(D) P lies inside C but outside E

Tangent to Ellipse-Slope Form

Q14. The equation of the tangents of the ellipse $9x^2+16y^2=144$ from the point (2, 3) are:

(A) $y=3, x+y=5$

(B) $y=3,\ x=2$

(C) $y=2, x=3$

(D) $y=3, x=5$

Tangent to Ellipse-Cartesian Form

Q15. The equation of the tangents drawn at the ends of the major axis of the ellipse $9x^2+5y^2-30y=0$ are :

(A) $y=\pm 3$ (B) $y=0, y=6$

(C) $x=\pm\sqrt{5}$ (D) None of these

Normals to Ellipse

Q16. The equation of normal at the point (0, 3) of the ellipse $9x^2+5y^2=45$ is :

(A) x-axis (B) y-axis

(C) $y+3=0$ (D) $y-3=0$

Q17. The number of maximum normals which can be drawn from a point to ellipse, is :

(A) 4 (B) 2

(C) 1 (D) 3

Q18. The value of λ, for which the line $2x-\frac{8}{3}\lambda y=-3$ is normal to the conic $x^2+\frac{y^2}{4}=1$, is :

(A) $-\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$

(C) –3 (D) $\pm\frac{\sqrt{3}}{2}$

Pair of tangent to Ellipse

Q19. The locus of the point of intersection of perpendicular tangents to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is:

(A) $x^2+y^2=a^2+b^2$ (B) $x^2-y^2=a^2+b^2$

(C) $x^2+y^2=a^2-b^2$ (D) $x^2-y^2=a^2-b^2$

Chord with a given Mid-Point

Q20. The equation of the chord of the ellipse $2x^2+5y^2=20$ which is bisected at the point (2, 1) is-

(A) $4x+5y+13=0$

(B) $4x+5y=13$

(C) $5x+4y+13=0$

(D) None of these

Exercise - 1A *Concept Builder*

MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Consider the ellipse

$$\frac{x^2}{f\left(k^2+2k+5\right)}+\frac{y^2}{f\left(k+11\right)}=1$$

and $f(x)$ is a positive decreasing function, then

(A) The set of values of k, for which the major axis is x-axis is (-3, 2)

(B) The set of values of k, for which the major axis is y-axis is $(-\infty,2)$

(C) The set of values of k, for which the major axis is y-axis is $(-\infty,-3)\cup(2,\infty)$

(C) The set of values of k, for which the major axis is y-axis is $(-3,\infty)$

Q2. If a number of ellipse (whose axes are x & y axes) be described having the same major axis 2a but a variable minor axis then the tangents at the ends of their latus rectum pass through fixed points which can be -

(A) (0,a) (B) (0,0)

(C) (0,–a) (D) (a,a)

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 2 to 4)

Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the parabola $y^2 = 2x$. They intersect at P and Q in the first and fourth quadrants respectively. Tangents to the ellipse at P and Q intersect the x- axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

Q3. The ratio of the areas of the triangles PQS and PQR, is

(A) 1 : 3 (B) 1 : 2

(C) 2 : 3 (D) 3 : 4

Q4. The area of quadrilateral PRQS, is

(A) $\frac{3\sqrt{15}}{2}$ (B) $\frac{15\sqrt{3}}{2}$

(C) $\frac{5\sqrt{3}}{2}$ (D) $\frac{5\sqrt{15}}{2}$

Q5. The equation of circle touching the parabola at upper end of its latus rectum and passing through its vertex, is -

(A) $2x^2 + 2y^2 - x - 2y = 0$

(B) $2x^2 + 2y^2 + 4x - \frac{9}{2}y = 0$

(C) $2x^2 + 2y^2 + x - 3y = 0$

(D) $2x^2 + 2y^2 - 7x + y = 0$

Q6. Match the following

Column I	Column II
A. An ellipse passing through the origin has its foci $(3,4)$ and $(6,8)$, then length of its minor axis is	p. 8
B. If PQ is focal chord of ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ which passes through $S=(3,0)$ and $PS=2$, then length of chord PQ is	q. $10\sqrt{2}$
C. If the line $y=x+K$ touches the ellipse $9x^2+16y^2=144$, then the difference of values of K is	r. 10
D. Sum of distances of a point on the ellipse $\frac{x^2}{9}+\frac{y^2}{16}=1$ from the foci	s. 12

The correct match is

(A) $A \to p, B \to q, C \to r, D \to s$

(B) $A \to q, B \to p, C \to s, D \to r$

(C) $A \to q, B \to r, C \to r, D \to p$

(D) $A \to s, B \to r, C \to p, D \to q$

SUBJECTIVE TYPE

Q7. Find the centre, the length of the axes, eccentricity and the foci of the ellipse $12x^2 + 4y^2 + 24x - 16y + 25 = 0$

Q8. Latus rectum of ellipse $4x^2 + 9y^2 - 8x - 36y + 4 = 0$ is ?

Q9. Find the position of the point $(4, -3)$ relative to the ellipse $5x^2 + 7y^2 = 140$.

Q10. For what value of λ does the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$.

Q11. Find the equation of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.

Q12. Find the equation of the tangents to the ellipse $x^2 + 16y^2 = 16$ each one of which makes an angle of $60°$ with the x-axis.

Q13. Find the equation of tangent to the ellipse $4x^2 + 9y^2 = 36$ at the point $(3, -2)$

Q14. Find the condition that the line $\ell x + my = n$ may be a normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. An ellipse has OB as semi minor axis, F and F' its focii and the angle FBF' is a right angle. Then the eccentricity of the ellipse is-

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{\sqrt{3}}$

Q2. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$, be the end points of the latus rectum of the ellipse $x^2+4y^2=4$. The equations of parabolas with latus rectum PQ are -

(A) $x^2 + 2\sqrt{3}\,y = 3 + \sqrt{3}$

(B) $x^2 - 2\sqrt{3}\,y = 3 + \sqrt{3}$

(C) $x^2 + 3\sqrt{3}\,y = 3 - \sqrt{3}$

(D) $x^2 - 3\sqrt{3}\,y = 3 - \sqrt{3}$

Q3. The equation, $2x^2 + 3y^2 - 8x - 18y + 35 = K$ represents

(A) No locus if $K > 0$

(B) An ellipse if $K < 0$

(C) A point if $K = 0$

(D) A hyperbola if $K > 0$

Q4. A circle has the same centre as an ellipse & passes through the foci F_1 & F_2 of the ellipse, such that the two curves intersect in 4 points. Let 'P' be any one of their point of intersection. If the major axis of the ellipse is 17 & the area of the triangle PF_1F_2 is 30, then the distance between the foci is :

(A) 11 (B) 12

(C)13 (D) none

Q5. The positive difference between the length of the latus rectum of $3y = x^2 + 4x - 9$ and $x^2 + 4y^2 - 6x + 16y = 24$ is-

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Q6. If α & β are the eccentric angles of the extremities of a focal chord of an standard ellipse, then the eccentricity of the ellipse is :

(A) $\frac{\cos\alpha + \cos\beta}{\cos(\alpha+\beta)}$ (B) $\frac{\sin\alpha - \sin\beta}{\sin(\alpha-\beta)}$

(C) $\frac{\cos\alpha - \cos\beta}{\cos(\alpha-\beta)}$ (D) $\frac{\sin\alpha + \sin\beta}{\sin(\alpha+\beta)}$

Q7. If P is any point on ellipse with foci S_1 & S_2 and eccentricity is $\frac{1}{2}$ such that $\angle PS_1S_2 = \alpha$, $\angle PS_2S_1 = \beta$, $\angle S_1PS_2 = \gamma$, then $\cot\frac{\alpha}{2}$, $\cot\frac{\gamma}{2}$, $\cot\frac{\beta}{2}$ are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) ***NOT*** A.P., G.P. & H.P.

Q8. The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$

Q9. Imagine that you have two thumbtacks placed at two points, A and B. If the ends of a fixed length of string are fastened to the thumtacks and the string is drawn taut with a pencil, the path traced by the pencil will be an ellipse. The best way to maximise the area surrounded by the ellipse with a fixed length of string occurs when

I the two points A and B have the maximum distance between them.

II two points A and B coincide.

III A and B are placed vertically.

IV The area is always same regardless of the location of A and B.

(A) I (B) II

(C) III (D) IV

Q10. The y-axis is the directrix of the ellipse with eccentricity $e = 1/2$ and the corresponding focus is at $(3, 0)$, equation to its auxiliary circle is

(A) $x^2 + y^2 - 8x + 12 = 0$

(B) $x^2 + y^2 - 8x - 12 = 0$

(C) $x^2 + y^2 - 8x + 9 = 0$

(D) $x^2 + y^2 = 4$

Q11. An ellipse is inscribed in a circle and a point within the circle is chosen at random. If the probability that this point lies outside the ellipse is 2/3 then the eccentricity of the ellipse is :

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{\sqrt{5}}{3}$

(C) $\frac{8}{9}$ (D) $\frac{2}{3}$

Q12. $x - 2y + 4 = 0$ is a common tangent to $y^2 = 4x$ & $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$. Then the value of b and the other common tangent are given by :

(A) $b = \sqrt{3}$; $x + 2y + 4 = 0$

(B) $b = 3$; $x + 2y + 4 = 0$

(C) $b = \sqrt{3}$; $x + 2y - 4 = 0$

(D) $b = \sqrt{3}$; $x - 2y - 4 = 0$

Q13. Equation of the common tangent to the ellipses, $\frac{x^2}{a^2+b^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{a^2+b^2} = 1$ is-

(A) $ay = bx + \sqrt{a^4 - a^2b^2 + b^4}$

(B) $by = ax - \sqrt{a^4 + a^2b^2 + b^4}$

(C) $ay = bx - \sqrt{a^4 + a^2b^2 + b^4}$

(D) $by = ax + \sqrt{a^4 - a^2b^2 + b^4}$

Q14. The minimum area of triangle formed by tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and coordinate axes-

(A) ab (B) $\frac{a^2+b^2}{2}$

(C) $\frac{(a+b)^2}{2}$ (D) $\frac{a^2+ab+b^2}{3}$

Q15. The normal at a variable point P on an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e′ such that :

(A) e′ is independent of e

(B) $e' = 1$

(C) $e' = e$

(D) $e' = 1/e$

Q16. The area of the rectangle formed by the perpendiculars from the centre of the standard ellipse to the tangent and normal at its point whose eccentric angle is $\pi/4$, is :

(A) $\frac{(a^2 - b^2)\,ab}{a^2 + b^2}$

(B) $\frac{(a^2 - b^2)}{(a^2 + b^2)ab}$

(C) $\frac{(a^2 - b^2)}{ab\,(a^2 + b^2)}$

(D) $\frac{a^2 + b^2}{(a^2 - b^2)\,ab}$

Q17. If $\tan\theta_1 \tan\theta_2 = -\frac{a^2}{b^2}$ then the chord joining two points θ_1 & θ_2 on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ will subtend a right angle at -

(A) focus

(B) centre

(C) end of the major axis

(D) end of the minor axis

Q18. The locus of the middle point of chords of an ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ passing through P(0, 5) is another ellipse E. The coordinates of the foci of the ellipse E, is

(A) $\left(0, \frac{3}{5}\right)$ and $\left(0, \frac{-3}{5}\right)$

(B) (0, – 4) and (0, 1)

(C) (0, 4) and (0, 1)

(D) $\left(0, \frac{11}{2}\right)$ and $\left(0, \frac{-1}{2}\right)$

Q19. Point 'O' is the centre of the ellipse with major axis AB & minor axis CD. Point F is one focus of the ellipse. If OF = 6 & the diameter of the inscribed circle of triangle OCF is 2, then find the product (AB) (CD).

(A) 65 (B) 66

(C) 56 (D) 60

Exercise - 2A — *Brain Booster*

MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Extremities of the latus rectum of the ellipses $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b) having a given major axis 2a lies on-

(A) $x^2 = a(a - y)$

(B) $x^2 = a(a + y)$

(C) $y^2 = a(a + x)$

(D) $y^2 = a(a - x)$

Q2. If ax + by = 1, $cx^2 + dy^2 = 1$ have only one solution, then:

(A) $\frac{a^2}{c}+\frac{b^2}{d}=1$

(B) x = a/c

(C) y = b/d

(D) $a^2 + b^2 = 1$

Q3. Which of the following is an equation of the ellipse with centre (–2,1), major axis running from (–2,6) to (–2,–4) and focus at (–2,5) ?

(A) $\frac{(x-2)^2}{25}+\frac{(y+1)^2}{16}=1$

(B) $\frac{(x+2)^2}{25}+\frac{(y-1)^2}{9}=1$

(C) $\frac{(x-2)^2}{9}+\frac{(y+1)^2}{25}=1$

(D) $\frac{(x+2)^2}{9}+\frac{(y-1)^2}{25}=1$

Q4. Which of the following statement(s) is/are correct for the ellipse of **Question No. 3 (Above Quesion)** ?

(A) auxiliary circle is $(x + 2)^2 + (y - 1)^2 = 25$

(B) director circle is $(x + 2)^2 + (y - 1)^2 = 34$

(C) Latus rectum $= \frac{18}{5}$

(D) eccentricity $= \frac{4}{5}$

Q5. A tangent to the ellipse $4x^2 + 9y^2 = 36$ is cut by the tangent at the extremities of the major axis at T and T′. The circle on TT′ as diameter passes through the point -

(A) $(-\sqrt{5}, 0)$ (B) $(\sqrt{5}, 0)$

(C) (0, 0) (D) (3, 2)

Q6. Consider the ellipse $\frac{x^2}{\tan^2\alpha}+\frac{y^2}{\sec^2\alpha}=1$ where $\alpha \in (0,\pi/2)$. Which of the following quantities would vary as α varies ?

(A) Degree of flatness

(B) Ordinate of the vertex

(C) Coordinates of the foci

(D) Length of the latus rectum

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Passage # 1 (Q. 7 to 9)

For the ellipse $\frac{(x+y-1)^2}{9}+\frac{(x-y+2)^2}{4}=1$

Q7. The equation of minor axis is -

(A) $x+y-1=0$

(B) $x-y+2=0$

(C) $x=\frac{1}{2}$

(D) $y=\frac{3}{2}$

Q8. Ends of minor axis are -

(A) $\left(\frac{1}{2},\frac{1}{2}\right),\left(\frac{-3}{2},\frac{5}{2}\right)$

(B) (1, 3) (–2, 0)

(C) (2, 4), (0, 2)

(D) None of these

Q9. Length of major axis and minor axis are-

(A) $3\sqrt{2}, 2\sqrt{2}$ (B) 6, 4

(C) 9, 4 (D) 3, 2

Paragraph # 2 (Ques. 10 to 13)

Let the two foci of an ellipse be (– 1, 0) and (3, 4) and the foot of perpendicular from the focus (3, 4) upon a tangent to the ellipse be (4, 6).

Q10. The foot of perpendicular from the focus (– 1, 0) upon the same tangent to the ellipse is

(A) $\left(\frac{12}{5},\frac{34}{5}\right)$ (B) $\left(\frac{7}{3},\frac{11}{3}\right)$

(C) $\left(2,\frac{17}{4}\right)$ (D) (– 1, 2)

Q11. The equation of auxiliary circle of the ellipse is

(A) $x^2+y^2-2x-4y-5=0$

(B) $x^2+y^2-2x-4y-20=0$

(C) $x^2+y^2+2x+4y-20=0$

(D) $x^2+y^2+2x+4y-5=0$

Q12. The length of semi-minor axis of the ellipse is

(A) 1 (B) $2\sqrt{2}$

(C) $\sqrt{17}$ (D) $\sqrt{19}$

Q13. The equations of directrices of the ellipse are

(A) $x-y+2=0,\ x-y-5=0$

(B) $x+y-\frac{21}{2}=0, x+y+\frac{17}{2}=0$

(C) $x-y+\frac{3}{2}=0, x-y-\frac{5}{2}=0$

(D) $x+y-\frac{31}{2}=0, x+y+\frac{19}{2}=0$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14. **Column-I** **Column-II**

(A) The eccentricity of the ellipse which meets the straight line $2x - 3y = 6$ on the X-axis and the straight line $4x + 5y = 20$ on the Y-axis and whose principal axes lie along the coordinate axes, is (P) $\frac{1}{2}$

(B) A bar of length 20 units moves with its ends on two fixed straight lines at right angles. A point P marked on the bar at a distance of 8 units from one end describes a conic whose eccentricity is (Q) $\frac{1}{\sqrt{2}}$

(C) If one extremity of the minor axis of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the foci form an equilateral triangle, then its eccentricity, is (R) $\frac{\sqrt{5}}{3}$

(D) There are exactly two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose distance from the centre of the ellipse are greatest and equal to $\sqrt{\frac{a^2+2b^2}{2}}$. Eccentricity of this ellipse is equal to (S) $\frac{\sqrt{7}}{4}$

Q15. **Column I** **Column II**

(A) Stick of length 10 meter slides on co-ordinate axes, then locus of a point dividing this stick reckoning from x axis in the ratio of 6 : 4 is a curve whose eccentricity is e, then e is equal to (P) $\sqrt{6}$

(B) AA′ is major axis of an ellipse $3x^2 + 2y^2 + 6x - 4y - 1 = 0$ & P is a variable point on it then greatest area of triangle APA′ is (Q) $2\sqrt{7}$

(C) Distance between foci of the curve represented by the equation $x = 1 + 4\cos\theta$, $y = 2 + 3\sin\theta$ is (R) $\frac{128}{3}$

(D) Tangents are Drawn to the ellipse $\frac{x^2}{16} + \frac{y^2}{7} = 1$ at end points of latus-rectum. The area of quadrilateral so formed is (S) $\frac{\sqrt{5}}{3}$

Q16. Consider an ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ with centre C and a point P on it with eccentric angle $\frac{\pi}{4}$. Normal drawn at P intersects the major and minor axes in A and B respectively. N_1 and N_2 are the feet of the perpendiculars from the foci S_1 and S_2 respectively on the tangent at P and N is the foot of the perpendicular from the centre of the ellipse on the normal at P. Tangent at P intersects the axis of x at T.

Match the entries of **Column-I** with the entries of **Column-II**.

Column-I	**Column-II**
(A) (CA)(CT) is equal to	(P) 9
(B) (PN)(PB) is equal to	(Q) 16
(C) $(S_1N_1)(S_2N_2)$ is equal to	(R) 17
(D) $(S_1P)(S_2P)$ is equal to	(S) 25

NUMERICAL TYPE

Q17. Tangents drawn from the point P(2,3) to the circle $x^2 + y^2 - 8x + 6y + 1 = 0$ touch the circle at the points A and B. The circumcircle of the ΔPAB cuts the director circle of ellipse $\frac{(x+5)^2}{9}+\frac{(y-3)^2}{b^2}=1$ orthogonally.

Find the value of b^2.

Q18. Find the number of integral values of parameter 'a' for which three chords of the ellipse $\frac{x^2}{2a^2}+\frac{y^2}{a^2}=1$ (other than its diameter) passing through the point $P\left(11a,-\frac{a^2}{4}\right)$ are bisected by the parabola $y^2 = 4ax$.

Q19. Let P_i and P_i' be the feet of the perpendiculars drawn from foci S, S' on a tangent T_i to an ellipse whose length of semi-major axis is 20. If $\sum_{i=1}^{10}(SP_i)(S'P_i')=2560$, then find the value of 100e.

SUBJECTIVE TYPE

Q20. Find the equation of the ellipse having its centre at the point (2, –3), one focus at (3, –3) and one vertex at (4, –3)

Q21. If F_1 & F_2 are the feet of the perpendiculars from the foci S_1 & S_2 of an ellipse $\frac{x^2}{5}+\frac{y^2}{3}=1$ on the tangent at any point P on the ellipse, then $(S_1F_1)\,.\,(S_2F_2)$ is equal to

Q22. Find the equation to the locus of the middle point of the portion of the tangent to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ included between the co-ordinate axes.

Q23. Find the equation of the line on which the point of intersection of the tangents at the point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, and its corresponding point Q on the auxiliary circle meet.

Q24. Find the equation of the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$. Also find the length of the intercept of the tangent between the coordinate axes.

Q25. The tangent at the point α on a standard ellipse meets the auxiliary circle in two points which subtends a right angle at the centre. Show that the eccentricity of the ellipse is $(1 + \sin^2\alpha)^{-1/2}$.

Q26. If the normal at the point P(θ) to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$, intersects it again at the point Q(2θ), show that $\cos\theta = -(2/3)$.

Q27. If any two chords be drawn through two points on the major axis of an ellipse equidistant from the centre, show that $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}\cdot\tan\frac{\gamma}{2}\cdot\tan\frac{\delta}{2} = 1$, where $\alpha, \beta, \gamma, \delta$ are the eccentric angles of the extremities of the chords.

Q28. Prove that the equation to the circle, having double contact with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (with eccentricity e) at the ends of a latus rectum, is $x^2 + y^2 - 2ae^3x = a^2(1 - e^2 - e^4)$.

Q29. PG is the normal to a standard ellipse at P, G being on the major axis. GP is produced outwards to Q so that PQ = GP. Show that the locus of Q is an ellipse whose eccentricity is $\frac{a^2 - b^2}{a^2 + b^2}$.

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
D	A	B	C	A	B	A	D	D	D
11	12	13	14	15	16	17	18	19	20
C	C	D	A	B	B	A	D	A	B

Ex - 1A

1. A,C **2.** A,C **3.** C **4.** B **5.** D

6. C

7. Centre;(–1, 2), Length of major axis = 2a = $\sqrt{3}$, Length of minor axis = 2b = 1

foci $\left(-1, 2 \pm \frac{1}{\sqrt{2}}\right)$

8. $\frac{8}{3}$ **9.** Outside the ellipse **10.** $\lambda = \pm 5$

11. $y = \frac{1}{2}x \pm 2$ or $x - 2y \pm 4 = 0$ **12.** $y = \sqrt{3}x \pm 7$

13. $\frac{x}{3} - \frac{y}{2} = 1$ **14.** $1 = \frac{n^2}{(a^2 - b^2)^2}\left(\frac{a^2}{\ell^2} + \frac{b^2}{m^2}\right)$

Ex - 2									
1	2	3	4	5	6	7	8	9	10
A	B	C	C	A	D	A	B	B	A
11	12	13	14	15	16	17	18	19	
A	A	B	A	C	A	B	C	A	

Ex - 2A

1. A,B **2.** [A,B,C] **3.** D **4.** A,B,C,D **5.** [A,B]

6. A, B, D **7.** [A] **8.** [A] **9.** [A] **10.** A

11. B **12.** C **13.** D

14. (A) S, (B) R, (C) P, (D) Q **15.** A $\rightarrow$ S; B $\rightarrow$ P; C $\rightarrow$ Q; D $\rightarrow$ R

16. (A) Q; (B) S; (C) P; (D) R **17.** 54 **18.** B

19. 60 **20.** $3x^2 + 4y^2 - 12x + 24y + 36 = 0$

21. 3 **22.** $3x^2 + 4y^2 = 4x^2y^2$ **23.** $y = 0$ **24.** $\frac{14}{\sqrt{3}}$